a

spin configuration $\{\hat{n}_i\}$ on the lattice (S^2 field)

SO(3)-equivariant encoder (e3nn, fixed features + cell network)

latent: irreps $l = 0..3$ (4 channels each) global $z_g \sim N(0, I)$, cell $z_c \sim N(z_g, \tau_l^2)$ (top-down prior)

decoder $\lambda_{lm}(z)$, $l = 1..6$ (unrestricted) $E(\hat{n}) = -\sum \lambda_{lm} Y_{lm}(\hat{n})$ $p(\hat{n}) \propto e^{-E}$

likelihood per site: $\lambda \cdot Y(\hat{n}_i) - \log Z(\lambda)$, $\log Z$ by Lebedev quadrature; KL to the hierarchical prior

b

$l = 1$: one cap (vMF, $\kappa = 20$)

dipole (polar order)

$l = 2$: two caps (Maier–Saupe, $a_2 = 4$)

nematic order

$l = 3$: three coplanar caps (D_{3h}; widths 0.1 in / 0.3 out of plane)

octupolar order